

Week 5 – Healthcare Data Cleaning and Modeling

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1. Introduction:

This project uses Power BI to analyze healthcare data related to patients, admissions, billing, insurance, and medical conditions through an interactive dashboard.

2. Data Loading:

The healthcare dataset was imported into Power BI using **Get Data → CSV** and prepared for further analysis.

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider
Laura Myers	61	Male	B+	Obesity	12 March 2021	Kyle Williams MD	Brennan Bentley, and Hunter	Cigna
William Rogers	84	Male	A-	Asthma	31 August 2020	Heather Petty	Robinson-Ritter	Cigna
Jason Smith	74	Male	A-	Obesity	03 September 2022	Diana Martinez	Mcknight Perry Rivera, and	Cigna
Rachel Bush	61	Female	O-	Hypertension	27 March 2020	Adrian Vaughn	Lopez Ingram, and Barber	Cigna
Nicole Perry	83	Female	O-	Asthma	08 February 2020	Susan Miranda	Inc Soto	Cigna
Donald Simon	66	Male	B+	Hypertension	27 November 2023	James Hendrix	Morris and Sons	Cigna
Carlos Edwards	73	Male	B-	Hypertension	07 September 2022	Nichole Salas	Vaughn-Williamson	Cigna
Mrs. Jennifer Chandler	77	Female	B-	Asthma	27 November 2020	Jessica Schroeder	Willis PLC	Cigna
Stephen Holt	84	Male	O+	Cancer	23 June 2023	Richard Bailey	Chavez, and Smith Hawkins	Cigna
Jose Rivas	81	Female	AB-	Asthma	17 February 2022	Shannon Meyer	Durham, Burch and Webb	Cigna
Jamie Kennedy	62	Male	O+	Diabetes	07 January 2020	Adam Bradley	Frye Inc	Cigna
Alexander Jordan	84	Male	O-	Hypertension	23 April 2023	Mario Gonzales	Garcia-Jacobs	Cigna
Holly Harrell	71	Male	O-	Diabetes	13 June 2022	Kimberly Campbell	Santiago Sons and	Cigna
Angela Leonard	80	Male	AB-	Obesity	20 January 2021	James Goodwin	Kline, Davis Thomas and	Cigna
Brent Gibson	63	Male	O+	Asthma	01 November 2019	Christopher Higgins	Lee, Martinez Hart and	Cigna
Lori Hughes	73	Male	B-	Hypertension	06 March 2020	Janice Hancock	LLC Barnes	Cigna
Mr. Nicholas Garcia	64	Female	A-	Diabetes	03 July 2020	Mathew Williams	Lopez, and Conner Evans	Cigna
Sharon Castro	81	Female	B-	Diabetes	21 March 2024	Holly Harris	and Williams Day, Tyler	Cigna
Christopher Clark	70	Female	AB-	Arthritis	25 July 2019	Jordan Haney	and Greene, Love Gonzalez	Cigna
Benjamin Brown	62	Female	B-	Hypertension	03 March 2021	Samantha Bradford	Rogers PLC	Cigna
Alexander Miller	70	Female	AB-	Asthma	05 February 2023	Michael Buchanan	and Thomas Lopez Lamb,	Cigna
Michael Gillespie	85	Female	O+	Asthma	23 January 2024	Anthony Robertson	Gonzalez Roman, Orozco and	Cigna

3. Data Cleaning:

Power Query was used to handle missing values, correct data types, remove duplicates, and standardize the data.

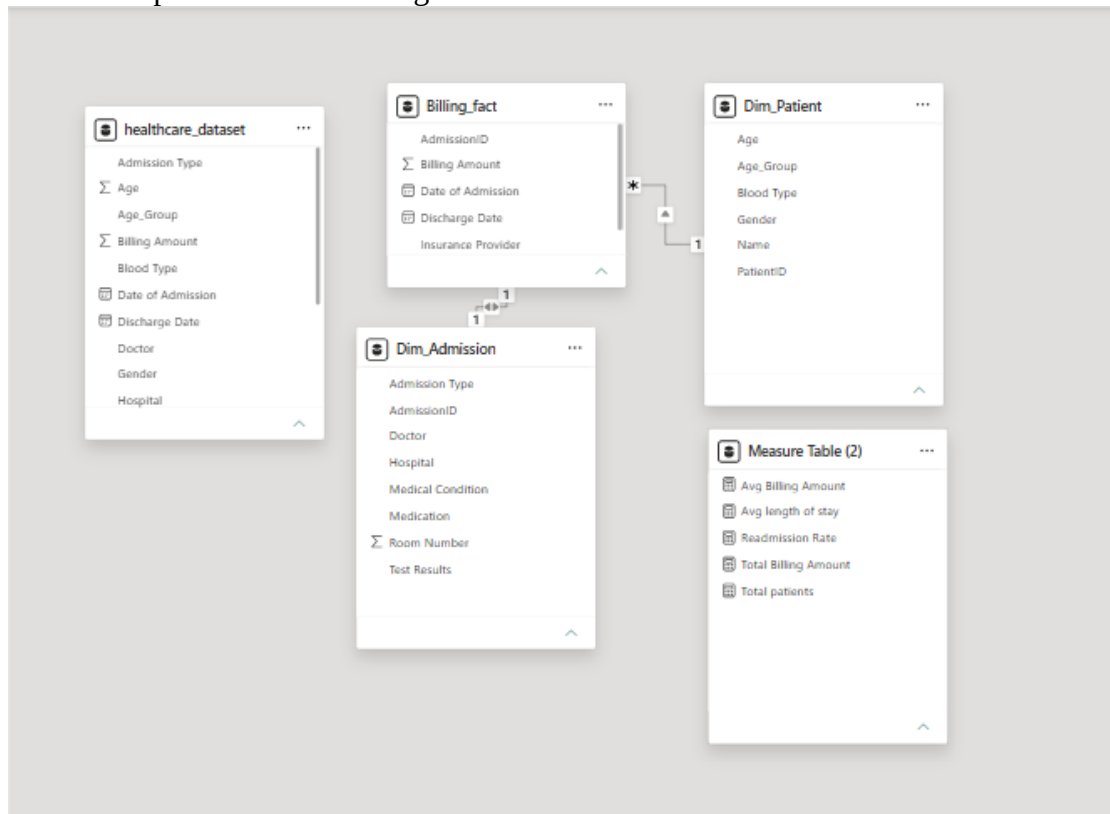
Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication
Franco-Campos	Cigna	11803.4882606488	263	Urgent	01 December 2023	Lipitor
Woods-Davidson	Cigna	7039.17535380228	399	Urgent	24 May 2022	Lipitor
Ferguson, Maldonado Johnson and	Cigna	13020.7708986913	163	Urgent	10 March 2020	Lipitor
and Wilson Santiago, Moore	Cigna	21094.9576214969	332	Urgent	18 November 2020	Lipitor
Mckee LLC	Cigna	20332.9439179936	499	Urgent	02 March 2023	Lipitor
Elliott-Sutton	Cigna	2635.89912036468	303	Urgent	26 April 2020	Lipitor
Hill-Rivera	Cigna	45793.9109950531	430	Urgent	29 August 2021	Lipitor
Gray Group	Cigna	31271.2388574917	215	Urgent	30 August 2023	Lipitor
Hernandez-Schroeder	Cigna	31432.5798493318	109	Urgent	04 November 2021	Lipitor
LLC Lee	Cigna	19369.7910720276	262	Urgent	19 May 2024	Lipitor
Mendoza Foley Williams, and	Cigna	13682.908107363	233	Urgent	08 August 2021	Lipitor
Stewart and Sanchez Lawrence,	Cigna	3285.61276705679	317	Urgent	07 September 2020	Lipitor
Henderson Ltd	Cigna	33307.4987045071	489	Urgent	21 May 2020	Lipitor
Anderson-Melendez	Cigna	23508.969698794	306	Urgent	01 September 2019	Lipitor
and Richard White, Sullivan	Cigna	30152.1181064124	378	Urgent	09 November 2019	Lipitor
Moore Inc	Cigna	29798.8486753847	453	Urgent	08 January 2020	Lipitor
and Bates Smith Wade,	Cigna	23782.6454298565	391	Urgent	14 May 2024	Lipitor
Rogers-Gutierrez	Cigna	17312.7440545881	315	Urgent	28 August 2022	Lipitor
PLC Leblanc	Cigna	29334.9740939415	374	Urgent	20 December 2022	Lipitor
Oconnell Group	Cigna	39514.7306710606	292	Urgent	16 May 2021	Lipitor
West Inc	Cigna	27869.4761996175	303	Urgent	07 May 2024	Lipitor
Conner Fitzgerald and Richardson,	Cigna	35966.6378861008	478	Urgent	29 June 2019	Lipitor

4. Age Group Creation:

An **Age_Group** column was created to classify patients into **Child, Young Adult, Adult, and Senior** categories.

5. Data Modeling:

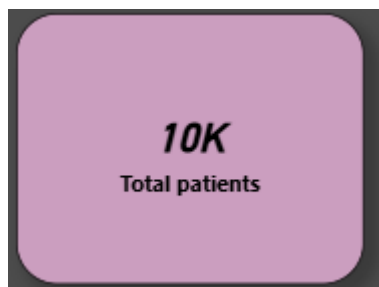
The data was organized into **Dim_Patient, Dim_Admission, and Billing** tables. Relationships were created using **PatientID** and **AdmissionID**.



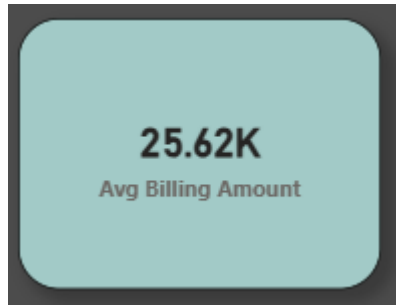
6. Dashboard KPIs:

The dashboard displays:

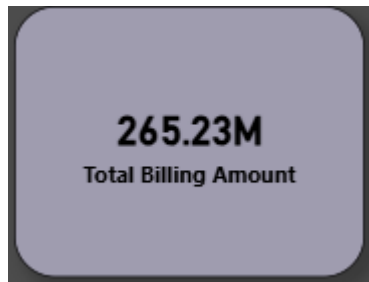
- **Total Patients – 10K**



- **Average Billing Amount – 25.62K**



- **Total Billing Amount – 265.23M**



7. Dashboard Visuals:

The dashboard includes:

- Date of Admission slicer
- Admission Type filter
- Insurance Provider filter
- Readmission Rate by Medical Condition
- Patients and Billing by Year
- Gender and Age Group table
- Billing Amount by Insurance Provider
- Patients by Age Group

8. Key Insights:

- The dashboard contains around **10K patients**.
- Average billing is approximately **25.62K**.
- Total billing reaches approximately **265.23M**.
- Billing and patient distribution can be analyzed by gender, age group, year, and insurance provider.
- Medical conditions show different readmission rates.

9. Conclusion:

The project demonstrates how Power BI can be used for healthcare data cleaning, modeling, visualization, and interactive analysis of patient and billing information.